

Full Length Article

Effect of hydrothermal pretreatment on two-stage anaerobic digestion of food waste and *Enteromorpha*: Digestion performance, bioenergy efficiency, and microbial community dynamics

Yufang Wei^{a,b}, Jun Gao^c, Zhengui Shi^d, Xiuji Li^{a,*}, Wenchao Ma^e, Hairong Yuan^{a,*}

^a State Key Laboratory of Chemical Resource Engineering, Department of Environmental Science and Engineering, Beijing University of Chemical Technology, 15 Beisanhuan East Road, Chaoyang District, Beijing 100029, PR China

^b State Environmental Protection Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences, Beijing 100012, PR China

^c Qingdao Greensum Ecology Co., Ltd., No. 6 Youyun Road, Laoshan District, Qingdao, Shandong 266100, PR China

^d Aerospace Kaitian Environmental Technology Co., Changsha, Hunan 410100, PR China

^e Tianjin Key Lab of Biomass Waste Utilization, School of Environmental Science and Engineering, Key Laboratory of Efficient Utilization of Low and Medium Grade Energy (Ministry of Education), Tianjin University, Tianjin 300350, PR China

ARTICLE INFO

Keywords:

Hydrothermal pretreatment
Two-stage anaerobic digestion
Enteromorpha
Food waste
Bioenergy efficiency
Microbial community structure

ABSTRACT

The effects of hydrothermal pretreatment (HTP) on the performance, bioenergy efficiency, and microbial community dynamics of two-stage anaerobic digestion (AD) of food waste (FW) and *Enteromorpha* were analyzed. The results showed that hydrogen production and total volatile fatty acids (VFAs) concentration in the dark fermentation (DF) stage and cumulative biomethane yield in the methanogenic stage exhibited an increasing trend in response to increasing HTP temperature. The highest hydrogen and biomethane yields of 62.7 ± 0.3 and 590.6 ± 8.0 mL·gVS⁻¹ were obtained at HTP temperature of 140 °C, which were 242.6% and 11.8% higher than those in two-stage AD with untreated substrates (designated as RTCK), respectively. Simultaneously, bioenergy generation yield (BGY) of 21.18 ± 0.44 – 24.21 ± 0.32 kJ·gVS⁻¹ in two-stage AD achieved bioenergy conversion efficiency (BCE) of 71.69–80.37% and process energy efficiency (PEE) of 22.31–39.01%, which showed 14.27–16.91% and 24.83–31.15% improvement compared to one-stage AD, respectively. HTP significantly altered the bacterial communities, resulting in the enrichment of butyric- and acetic-type fermentation bacteria and the inhibition of propionate-type fermentation bacteria. This study demonstrated that HTP could improve the performance and achieve a higher BGY in two-stage AD.

1. Introduction

Bioenergy generation technologies, representing promising alternative ways to convert biomass waste into renewable and sustainable green bioenergy, are being developed vigorously under the policy framework for carbon peaking and carbon neutrality in China. Anaerobic digestion (AD) is a sequence of biological processes involving multiple equilibria between microbial metabolism that can recover bioenergy in the form of hydrogen (H₂) and/or biomethane (CH₄) and reduce the environmental impact [1,2]. AD involves four phases: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. Each phase requires specific bacteria and archaea. In traditional AD (one-stage AD), the microorganisms are kept together in one reactor, resulting in

problems including process instability, accumulation of volatile fatty acids (VFAs), and poor impact resistance [3,4]. Meanwhile, low bio-methane yield is a developmental bottleneck for one-stage AD [5]. This is because the microorganisms involved in each phase widely differ in physiology, nutritional needs, growth kinetics, and environmental conditions [3,4]. Two-stage AD can overcome these inherent complications by physically separating the reactors into a dark fermentation (DF) reactor for hydrogen generation, and a methanogenic reactor for biomethane generation, enabling the optimal conditions for specific microorganisms [3,6].

Food waste (FW), with an annual production of 195 million tons in China, comprises 41–62% carbohydrates, 15–25% proteins, and 13–30% lipids [7]. *Enteromorpha*, which forms green tides during

* Corresponding authors.

E-mail addresses: xjli@mail.buct.edu.cn (X. Li), yuanhairong75@163.com (H. Yuan).

<https://doi.org/10.1016/j.fuel.2022.123639>

Received 5 January 2022; Received in revised form 11 February 2022; Accepted 15 February 2022

Available online 24 February 2022

0016-2361/© 2022 Elsevier Ltd. All rights reserved.

eutrophication, is structurally configured with 60–65% carbohydrates, 16–30% proteins, 4–5% lipids, low cellulose, and negligible lignin contents [8]. Millions of tons of *Enteromorpha* attacked Qingdao, Shandong Province for the 15th consecutive year [9]. Improper management and disposal of FW or *Enteromorpha* result in social, economic, and environmental problems [2,10]. In this scenario, the viability and utilization of FW or *Enteromorpha* for bioenergy production via AD have been explored [8,11–13]. Nevertheless, the high lipid content in FW and structurally recalcitrant compounds in *Enteromorpha* results in the accumulation of VFAs and hinder the microbial degradation during AD, resulting in instability or irreversible failure of one-stage AD [7,8,11]. Thus, it is necessary to pretreat these substrates before AD to obtain a higher bioenergy efficiency and productivity.

Hydrothermal pretreatment (HTP) of substrates has been identified as the most promising technique for achieving satisfactory digestion performance, because of its high efficiency, chemical-free nature, and environment-friendly nature [1,14,15]. In this regard, the hydrogen and biomethane yields of 43.0 and 511.6 mL·gVS⁻¹ were obtained, respectively, in two-stage AD of FW with HTP at 140 °C, resulting in 31.7% improvement in bioenergy conversion efficiency (BCE) [16]. Similarly, HTP at 220 °C resulted in hydrogen and biomethane yields of 23.3 and 877.5 mL·gVS⁻¹, respectively, from FW through one-stage AD [7]. Yin et al. [17] reported that HTP enhances total VFA production by obtaining a total VFA concentration of 34.1 g·L⁻¹ from FW at an HTP temperature of 160 °C. However, high-temperature HTP (above 150 °C) consumes considerable energy and inevitably forms melanoidins through the Maillard reaction, which can damage the BCE and jeopardize the techno-economics of the overall process [14,18]. Consequently, the investigation of the effect of HTP at low temperatures (below 150 °C) on the co-digestion of FW and *Enteromorpha* is valuable and meaningful; however, little research has been conducted on this topic.

To date, most studies have investigated the possibility of combining the recovery of both hydrogen and biomethane through two-stage AD [4,15,19]. Meanwhile, studies on effect of HTP on the bioenergy efficiency in the two-stage AD process, as well as comparison with one-stage AD are limited. Coordinated microbial activities and biochemical reactions result in successful AD [20]. Knowledge about microbial community dynamics in one-stage AD is well established, including the distribution of microbes and their shifts to HTP conditions [21,22]. A few studies on microbial properties have been conducted in two-stage AD of FW with HTP; however, the succession of microbial communities during two-stage AD of FW and *Enteromorpha* with low-temperature HTP has not been reported thus far. Furthermore, the relationship between bioenergy performance and microbial communities has rarely been reported.

The objectives of this study were to: (1) assess the performance in terms of hydrogen and biomethane production yields in two-stage AD of FW and *Enteromorpha* with low-temperature HTP; (2) evaluate the bioenergy efficiency of one-stage AD and two-stage AD with HTP; and (3) investigate the structure, dynamics, and functional shift of the microbial community during two-stage AD with HTP.

2. Materials and methods

2.1. Substrates and inoculum

FW was obtained from a student canteen at Beijing University of Chemical Technology, Beijing, China. After the removal of impurities such as bones, plastic bags, and plastic cups, the FW was milled using a food grinder (Element40, BECBAS, China) and stored at -20 °C in a laboratory cryobiology refrigerator (DW-40L508, Haier, China) for experimental use. Fresh *Enteromorpha* was collected from the seaside of Laoshan District, Qingdao, Shandong, China. The fresh *Enteromorpha* was air-dried in an open field, and then pulverized to approximately 0.25 mm using a hammer mill (YSW-180, Zhengde, China) after the removal of sands and plastic bags. The inoculum used in the study was

the liquid fraction of digestate (LFD) obtained from the mesophilic digestion reactor of the Beijing Dongcun Comprehensive Municipal Solid Waste Treatment Plant located in Tongzhou District, Beijing, China, which has been stably operated for more than three years using FW as a mono-substrate. Oyster shells, collected from Binhai New Area, Tianjin, China, were pulverized to less than 0.21 mm using a hammer mill (YSW-180, Zhengde, China). The characteristics of the FW, *Enteromorpha*, inoculum, and oyster shells are listed in Table 1.

2.2. Experimental setup

2.2.1. HTP process

The HTP process was conducted in a 1000 mL stainless-steel autoclave reactor (WZC-1000, WZDC, Beijing, China) with a Teflon lining inside. The total volatile solids (VS) concentration in the reactor was 30 g VS·L⁻¹. According to the pre-experimental results, the FW-to-*Enteromorpha* ratio was 3:1 on a VS basis. Based on the results of a previous study [23], the addition amount of oyster shells was 8% (w/w) of total dry weight of FW and *Enteromorpha*, to strengthen the system, stabilize pH, and improve the performance of DF [23–25]. The moisture content of the HTP process was set at 90%, which was adjusted by adding deionized water. The reactor was heated to the desired temperature (50, 80, 110, or 140 °C) and maintained for 20 min. The HTP time was precisely measured, starting at the time the reactor reached the set temperature. After the HTP process, the reactor was cooled to 25 °C using tap water, and the pretreated substrates were removed for subsequent analyses and digestion trials.

2.2.2. Experimental set-up

Two-stage AD was operated using substrates with HTP at 50, 80, 110, and 140 °C, which were designated as RT50, RT80, RT110, and RT140, respectively. DF was conducted in a batch mode at 35 ± 1 °C using 500 mL glass bottles (300 mL working volume). The substrate-to-inoculum ratio was set to 6:1 on a VS basis. The DF stage lasted for seven days. The hydrogen production and performance of acidification during the entire DF stage were recorded. After the DF was completed, a part of the effluent was sampled for subsequent analyses and the residual effluents were then inoculated with the inoculum for methanogenic process. The organic loading rates of 30.0 and 60.0 gVS·L⁻¹ for substrates and inoculum, respectively, were set up for each methanogenic reactor. The substrate-to-inoculum ratio was 1:2 on a VS basis. The initial pH in the methanogenic reactor was maintained at 7.0 using 1.0 mol·L⁻¹ NaOH or HCl solutions [5]. Deionized water was subsequently added to a final volume of 300 mL in each methanogenic reactor to obtain a total volume

Table 1
Characteristics of substrates, inoculum, and oyster shells.

Items	FW	<i>Enteromorpha</i>	Inoculum	Oyster shell
Total solid (TS, %) ^a	26.8 ± 0.1	87.7 ± 0.3	10.4 ± 0.2	99.4 ± 0.1
Volatile solid (VS, %) ^a	24.8 ± 0.1	47.5 ± 0.2	5.1 ± 0.2	3.3 ± 0.1
Total carbon (TC, %) ^b	24.3	25.1	42.7	–
Total nitrogen (TN, %) ^b	1.7	2.7	3.2	–
Total hydrogen (TH, %) ^b	4.7	4.0	6.4	–
Total sulfur (TSu, %) ^b	2.9	0.9	0.1	–
Total oxygen (TO, %) ^{b, c}	60.8	59.6	1.7	–
Ash (T _{Ash} , %) ^b	5.6	7.7	45.9	50.5
C/N ^b	14.3	9.5	13.3	–
pH ^a	4.8 ± 0.1	–	7.3 ± 0.2	–

^cTO = 100%·TC·TH·TN·TSu·Tash.

– Not determined.

^a Content of fresh matter.

^b Content of dry matter.

of 500 mL. All methanogenic reactors were operated in triplicates at $35 \pm 1^\circ\text{C}$ for 50 days in a water tank with a temperature controller. The two-stage AD of untreated substrates was used as the control group (designated as RTCK). The hydrogen produced in the DF reactor and the biomethane produced in the methanogenic reactor were corrected by offsetting those produced from the inoculum in the blank control.

One-stage AD was performed in 500 mL glass bottles (working volume of 300 mL) using substrates with HTP at 50, 80, 110, and 140°C , which were designated as RO50, RO80, RO110, and RO140, respectively. The operation parameters including the organic loading rate, substrate-to-inoculum ratio, pH, operation temperature, and digestion time were the same as those in the methanogenic reactors in two-stage AD. One-stage AD of untreated substrates was used as the control group (designated as ROCK). One-stage AD and two-stage AD of the inoculum were conducted to deduct the background.

2.3. Analytical methods

2.3.1. Physicochemical properties analysis

Hydrogen production during the DF stage and biogas production during the methanogenic stage were monitored via the water-displacement method [23]. The biogas components (CH_4 , CO_2 , H_2 , and N_2) were detected using a gas chromatograph (GC, SP2100, Beifen, China) equipped with a $2\text{ m} \times 3\text{ mm}$ TDX-01 column and a thermal conductivity detector (TCD). The injector and detector temperatures were maintained at 150°C . Argon was used as the carrier gas. Biomethane production was corrected according to standard temperature (273.15 K) and pressure (100 kPa). Total solids (TS), VS, and alkalinity were determined according to standard methods published by the American Public Health Association. The total carbon, hydrogen, nitrogen and sulfur contents (represented as TC, TN, TH, and TSu, respectively) in the substrates and inoculum were analyzed using an elemental analyzer (Vario EL/micro cube, Germany). The ash content (T_{ASH}) was measured using a muffle furnace (SX2-2.5-12, Tianjin Tongda Experimental Electric Furnace Factory, Tianjin, China). The pH and oxidation–reduction potential (ORP) of the effluent samples were measured using a CHN868 pH meter (Thermo Orion, USA). The VFAs in the effluent samples were analyzed using a GC (GC-2014, Shimadzu, Japan) equipped with a flame ionization detector (FID) and a DBWAX123-7032 column. The temperature of the injector and detector was maintained at 250°C . A HANNA environmental testing photometer (HI83206, China) was used to determine $\text{NH}_4^+\text{-N}$ concentration.

2.3.2. Microbial community analyses

The profile response and functional shift of the microbial community in the effluent samples were elucidated using high-throughput 16S rRNA gene sequencing. The effluent samples for bacterial community analyses were collected from the DF process in RTCK on 12th hour, 24th hour, 4th days, and 7th days of the DF process, which were labeled as R1, R2, R3, and R4, respectively, and on the same time as RT140 (labeled as R5, R6, R7, and R8, respectively). The microbial community in the inoculum was also analyzed using a sample collected during inoculation (represented as R0). The effluent samples for archaeal community analyses were obtained at the end of the methanogenic stage of ROCK, RTCK, RO140, and RT140, as well as the sample collected during the inoculation. Microbial DNA extraction was performed using a FastDNA Spin Kit (MP Biomedicals, USA). Real-time polymerase chain reaction (PCR) amplification was performed using ABI GeneAmp® 9700 (ABI, USA). The primers used in this study were 338F (5'-ACTCCTACGGGAGG-CAGCAG-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') for bacteria and 524F_10_ext (5'-TGYCAGCCGCCGCGGTAA-3') and Arch 958R (5'-YCCGGCGTTGAVTCCAATT-3') for archaea. The purified products were sequenced using the Illumina MiSeq platform (Illumina, USA) according to standard protocols. The obtained information was analyzed to classify the microbial community structure using the I-Sanger cloud platform operated by Majorbio Bio-Pharm Technology Co. Ltd.

(Shanghai, China).

2.4. Bioenergy efficiency assessment

2.4.1. Process energy input

The total energy input (Q_{in} , $\text{kJ}\cdot\text{gVS}^{-1}$) for the process can be estimated using Eq. (1) [26].

$$Q_{\text{in}} = Q_{\text{HTP}} + Q_{\text{DF}} + Q_{\text{AD}} \quad (1)$$

where Q_{HTP} , Q_{DF} , and Q_{AD} are the energy inputs for the HTP, DF, and methanogenic stages, respectively.

The energy input for each stage is estimated using Eq. (2) [26].

$$Q_{\text{HTP}}/Q_{\text{DF}}/Q_{\text{AD}} = (V_{\text{Rea}} \cdot \rho \cdot C_p) \cdot x(T_{\text{Rea}} - T_{\text{Amb}}) \quad (2)$$

where V_{Rea} is the working volume (L) of the reactor used in each stage. The liquid fraction in the reactor was assumed to be water, with a density (ρ) and specific heat (C_p) of $1\text{ kg}\cdot\text{L}^{-1}$, and $4.2\text{ kJ}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$, respectively. T_{Rea} ($^\circ\text{C}$) and T_{Amb} ($^\circ\text{C}$) are the temperatures of the HTP and the ambient temperature (25°C), respectively. The minimum mass loss during each stage was neglected.

2.4.2. Bioenergy generation yield

The bioenergy generation yield (BGY , $\text{kJ}\cdot\text{gVS}^{-1}$), defined in Eq. (3), was calculated to evaluate the total bioenergy generation in each reactor, according to the methods reported by Yuan et al. [27].

$$BGY = V_{\text{H}_2} \cdot \hat{A} \cdot \text{HHV}_{\text{H}_2} + V_{\text{CH}_4} \cdot \hat{A} \cdot \text{HHV}_{\text{CH}_4} \quad (3)$$

where V_{H_2} and V_{CH_4} represent the volume yields ($\text{mL}\cdot\text{g VS}^{-1}$) of hydrogen and biomethane, respectively, at standard temperature (273.15°C) and pressure (101.325 kPa). HHV_{H_2} and HHV_{CH_4} refer to the high heating values of hydrogen and biomethane, which are 142.0 and $55.5\text{ kJ}\cdot\text{g}^{-1}$, respectively.

2.4.3. Bioenergy conversion efficiency and process energy efficiency

The BCE (%), defined as the percentage of BGY and theoretical bioenergy generation yield (TBY , $\text{kJ}\cdot\text{gVS}^{-1}$), was calculated using Eq. (4).

$$\text{BCE} = \frac{BGY}{TBY} \times 100\% \quad (4)$$

TBY (Eq. (5)) was calculated by the method described by Cao et al. [28], where X_{FW} and $X_{\text{Enteromorpha}}$ represent the VS percentages (%) of the FW and *Enteromorpha* used in this study, respectively. TBY_{FW} and $TBY_{\text{Enteromorpha}}$ refer to the TBY of FW and *Enteromorpha* used in this study, respectively, which were calculated according to Eq. (6).

$$TBY = X_{\text{FW}} \cdot \hat{A} \cdot \text{TBG}_{\text{FW}} + X_{\text{Enteromorpha}} \cdot \text{TBG}_{\text{Enteromorpha}} \quad (5)$$

$$\begin{aligned} TBY_{\text{substrate}} = & 0.3491wt_{\text{TC}} + 1.1783wt_{\text{TH}} + 0.1005wt_{\text{TSu}} - 0.0134wt_{\text{TO}} \\ & - 0.0151wt_{\text{TN}} - 0.0211wt_{\text{TAsh}} \end{aligned} \quad (6)$$

where wt_{C} , wt_{H} , wt_{S} , wt_{O} , wt_{N} , and wt_{TAsh} in Eq. (6) represent the weight percentages (%) of C, H, S, O, N, and ash in substrates listed in Table 1.

The process energy efficiency (PEE , %), defined as the percentage of BGY to Q_{in} , can be estimated using Eq. (7).

$$\text{PEE} = \frac{BGY}{Q_{\text{in}}} \times 100\% \quad (7)$$

2.5. Statistical analyses

All experiments were performed in triplicates, and the results were depicted as the mean value $\pm$ standard deviation (SD) using Microsoft Excel 2019. One-way analysis of variance followed by an LSD test was conducted at a probability level (p) less than 0.05, using the SPSS

software (version 16.0, IBM, USA), and the Origin software was used for data visualization.

3. Results and discussion

3.1. Performance of DF stage

3.1.1. Hydrogen production performance

Fig. 1 illustrates the hydrogen production performance during the DF stage using substrates treated at different HTP temperatures. During the entire DF period, the hourly hydrogen production of all reactors first declined and then increased. A similar trend was reported by Hou et al. [29], who operated two-stage AD of FW in batch mode. For the untreated sample (RTCK), the highest hydrogen production of $8.8 \pm 0.1 \text{ mL} \cdot \text{gVS}^{-1}$ was recorded at the 36th hour. When HTP temperature increased to 50, 80, 110, and 140°C , the highest peak hydrogen production gradually increased to 9.2 ± 0.5 , 33.5 ± 0.6 , 38.9 ± 1.5 , and $44.4 \pm 1.4 \text{ mL} \cdot \text{gVS}^{-1}$, resulting in 4.5, 280.7, 342.2, and 404.5% improvement, respectively, compared to that of RTCK ($8.8 \pm 0.1 \text{ mL} \cdot \text{gVS}^{-1}$) (Fig. 1A). The results indicate that a higher HTP temperature contributed to higher hydrogen production. In addition, the peak hydrogen production time advanced to 12th hour for RT80, RT110, and RT140. However, the highest peak hydrogen production for RT50 was the same as that of the control. The results confirmed that the HTP process resulted in the advancement of the highest hydrogen production when the HTP temperature was above 50°C , in accordance with the findings of He et al. [30].

At HTP temperatures of 50, 80, 110, and 140°C , the DF reactor obtained cumulative hydrogen yields of 31.5 ± 0.2 , 50.4 ± 0.2 , 58.5 ± 0.1 , and $62.7 \pm 0.3 \text{ mL} \cdot \text{gVS}^{-1}$, representing improvements of 72.1, 175.4, 219.7, and 242.6%, respectively, compared to the yield of RTCK ($18.3 \pm 0.3 \text{ mL} \cdot \text{gVS}^{-1}$) (Fig. 1B). During the HTP process, the soluble carbohydrate content increased with increasing HTP temperature, which could be easily utilized by microorganisms for hydrogen generation, resulting in a higher hydrogen yield [16]. Further increase in the HTP temperature resulted in an increase in the cumulative hydrogen yield, suggesting that *Enteromorpha* and FW could be good candidates to produce hydrogen through DF. Specifically, in the first 12 h, 33.2, 74.3, 79.6, and 84.5% of the total hydrogen yield were obtained for RT50, RT80, RT110, and RT140, respectively, exhibiting a considerable enhancement of 147.8–529.8% compared with the RTCK (13.4%). This indicated that the HTP temperature had a positive effect on hydrogen production. In addition, HTP at 140°C was the optimal condition for the DF of *Enteromorpha* and FW in this study.

3.1.2. Acidification performance

As shown in Fig. 2, the trends of the total concentration of VFAs and ethanol during the entire DF stage were clearly divided into two parts:

an upward trend from the start-up to day 3 or day 4, and followed by a downward trend until the end of DF process. This phenomenon corresponds with the results reported by Shi et al. [23] and Tian et al. [31]. The total VFA and ethanol concentrations recorded in RTCK were in the range of $5892.9\text{--}9481.4 \text{ mg} \cdot \text{L}^{-1}$. The highest total concentrations of VFA and ethanol obtained in RT50, RT80, RT110, and RT140 were 10633.7, 10771.4, 11925.1, and $12517.2 \text{ mg} \cdot \text{L}^{-1}$, respectively, resulting in the improvement of 12.2–32.0% when compared to RTCK. The results indicated that the HTP temperature had a positive impact on acidification performance, which could be explained by the improved VS removal rate (Table 2). Notably, hydrogen production almost ceased from the 2nd day in the DF stage, however, the total concentration of VFAs and ethanol increased before it decreased, warranting further study from the perspective of microorganisms. Furthermore, in comparison with that of RTCK ($7214.7 \text{ mg} \cdot \text{L}^{-1}$), the total concentration of VFAs and ethanol in reactors with HTP substrates at day 7 was reduced by 5.4–25.9%. A higher total concentration of VFAs and ethanol generally provides more intermediates for biomethane generation, ultimately resulting in higher biomethane production [12].

The percentages of VFAs components and ethanol varied throughout the duration of DF. No significant changes in the metabolic pathways were observed using pretreated substrates, as acetate and butyrate were the two dominant VFAs components, comprising 63.6–90.6% of the total concentration. These observations indicate that butyric- and acetic-type fermentation dominated during the DF stage, corresponding with the findings of Yin et al. [17] and Lin et al. [26]. As shown in Fig. 2, 38.5–59.2% of the VFAs concentration was butyric acid, and 16.0–49.0% of them was acetic acid. The proportion of butyric acid decreased with DF time; however, an opposite trend was observed for acetic acid. The concentrations of both butyric and acetic acids increased with increasing HTP temperature. A remarkable phenomenon of ethanol accumulation was observed in the first 3–4 days of DF stage, which was $364.2\text{--}958.0 \text{ mg} \cdot \text{L}^{-1}$ in RTCK, $756.0\text{--}2267.4 \text{ mg} \cdot \text{L}^{-1}$ in RT50, $1027.4\text{--}2371.5 \text{ mg} \cdot \text{L}^{-1}$ in RT80, $2092.4\text{--}2519.4 \text{ mg} \cdot \text{L}^{-1}$ in RT110, and $466.7\text{--}1109.4 \text{ mg} \cdot \text{L}^{-1}$ in RT140. The ethanol concentration fluctuated at a lower level in the following days, indicating a higher ethanol consumption rate than the production rate [32]. The isobutyric acid concentration was also enhanced via HTP, although its proportion was less than 12.6% throughout the DF stage. The propionic acid obtained in DF stage with and without HTP had a low concentration (less than $400 \text{ mg} \cdot \text{L}^{-1}$) and proportion (less than 6.3%), which were reported to be beneficial for the two-stage AD [6].

3.1.3. System stability and VS removal rate

The system stability parameters are closely related to the DF stage efficiency. In all reactors, the pH fluctuated between 5.23 ± 0.05 and 5.70 ± 0.04 (Table 2) and was in the desired pH range (5.00–6.00) for hydrogen generation by fermentation bacteria [33]. The stable pH was

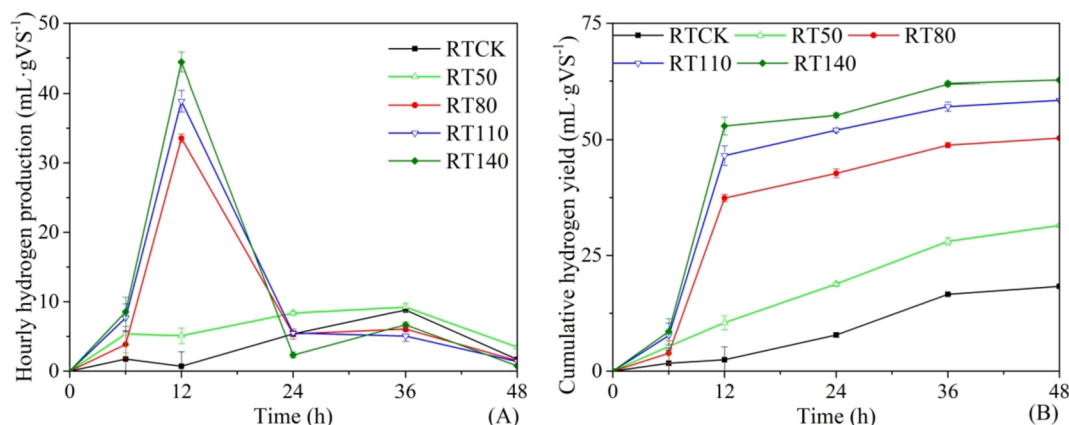


Fig. 1. Hourly hydrogen production (A) and cumulative hydrogen yield (B).

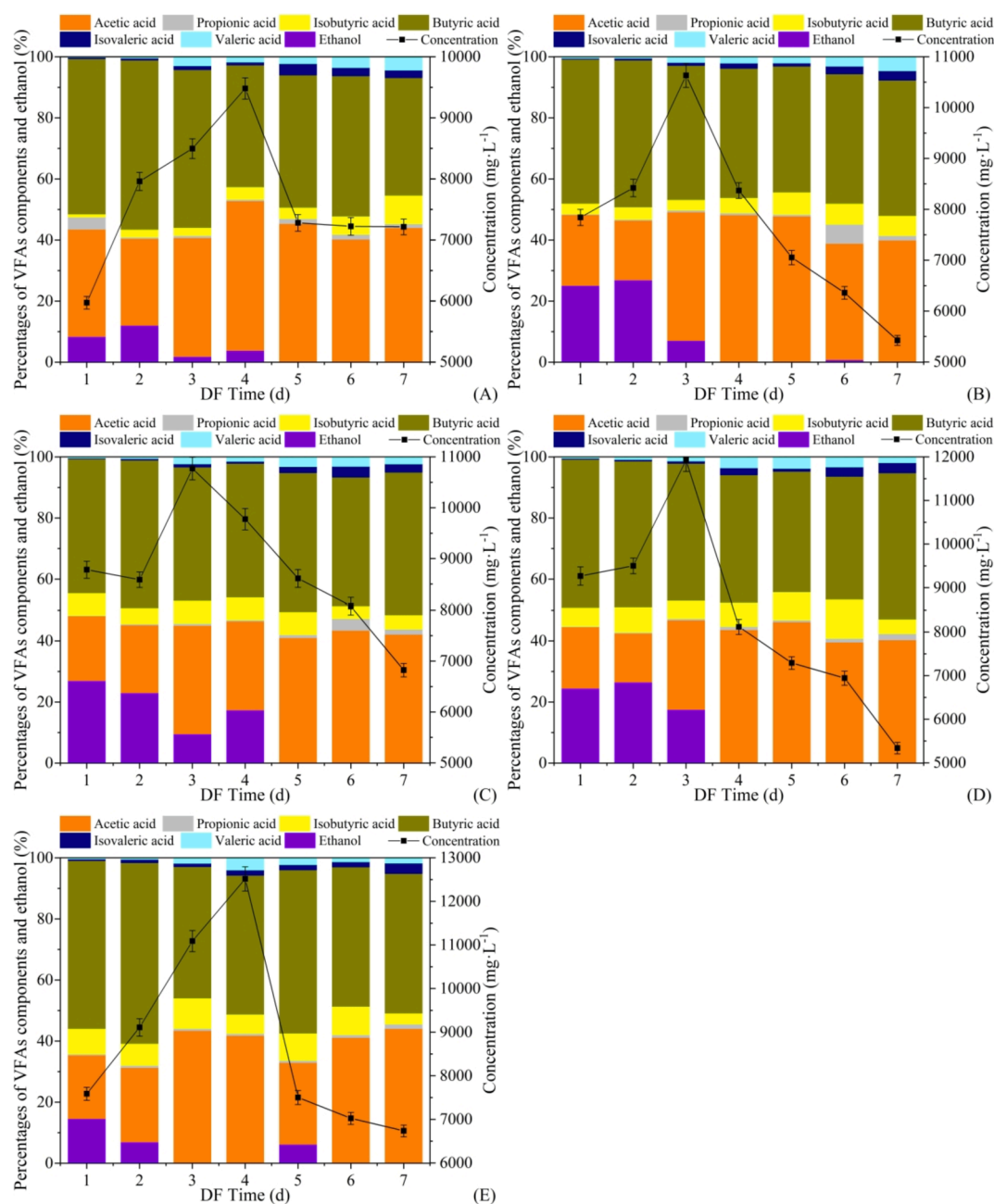


Fig. 2. The total concentration of VFAs and ethanol, and their percentages of RTCK (A), RT50 (B), RT80 (C), RT110 (D), and RT140 (E).

related to the addition of oyster shells, which have a high buffer capacity [23,24]. Furthermore, a continuous downward trend of pH with DF time (except for day 7) was observed in each reactor with and without HTP, which was consistent with the findings of Luo et al. [34]. ORP has a significant impact on the microbial community structure and metabolic pathways for hydrogen generation [35]. In RTCK, the ORP increased from $-(212.7 \pm 9.5)$ to $-(172.3 \pm 12.7)$ mV with time (Table 2). Similarly, an upward ORP trend from $-(211.0 \pm 8.2)$ to $-(167.0 \pm 10.1)$ mV was observed in DF reactors with HTP. ORP values above -278.0 mV contributed to stable butyric- and acetic-type fermentation [36], consistent with the analysis of VFAs. Typically, the degradation of proteins in substrates can release inorganic ammonia nitrogen into liquid as $\text{NH}_4^+\text{-N}$ or free ammonia (NH_3), and the latter is considered as an inhibitor for hydrogen production [111]. In this study, the $\text{NH}_4^+\text{-N}$

concentration observed in all DF reactors ranged from 305 ± 23 – 673 ± 18 $\text{mg}\cdot\text{L}^{-1}$, which is below the threshold (1500 $\text{mg}\cdot\text{L}^{-1}$) for methanogenesis [20]. Moreover, the calculated free ammonia concentration was below the threshold (80 $\text{mg}\cdot\text{L}^{-1}$), indicating the stable operation of the DF. Comparable alkalinity concentrations of 3766 ± 38 – 5400 ± 78 $\text{mg}\cdot\text{L}^{-1}$ were obtained in each DF reactor, suggesting suitable buffering capacity [6]. In this scenario, the VFA/alkalinity ratio exhibited a decreasing trend with time, suggesting that the system stability of the DF stage was greatly improved by HTP. The VS removal rate was positively correlated with duration (Table 2). With HTP, the VS removal rate was improved to (4.9 ± 0.2) – $(23.7 \pm 0.3)\%$, resulting in 58.0–172.2% improvement compared to that of RTCK. The VS removal rate generally increased with increasing HTP time (except for RT50), which corresponded with the previous discussion on acidification performance.

Table 2
System stability parameters and VS removal rate during DF stage.

Parameters	Reactor	DF time			
		1 day	3 days	5 days	7 days
pH	RTCK	5.70 ± 0.04	5.61 ± 0.04	5.46 ± 0.05	5.65 ± 0.03
		5.64 ± 0.02	5.63 ± 0.05	5.44 ± 0.13	5.60 ± 0.03
	RT50	5.62 ± 0.02	5.52 ± 0.08	5.23 ± 0.05	5.53 ± 0.08
		5.62 ± 0.01	5.44 ± 0.05	5.30 ± 0.11	5.62 ± 0.08
	RT80	5.62 ± 0.03	5.50 ± 0.02	5.48 ± 0.12	5.56 ± 0.24
	RT110				
	RT140				
ORP (mV)	RTCK	−(212.7 ± 9.5)	−(177.3 ± 7.1)	−(173.7 ± 9.8)	−(172.3 ± 12.7)
		−(211.0 ± 8.2)	−(193.3 ± 12.9)	−(176.7 ± 11.2)	−(170.7 ± 7.4)
	RT50	−(205.7 ± 8.7)	−(203.0 ± 5.6)	−(146.3 ± 13.1)	−(169.3 ± 9.5)
		−(192.7 ± 4.6)	−(176.3 ± 8.5)	−(157.3 ± 7.7)	−(167.0 ± 10.1)
	RT80	−(191.0 ± 6.7)	−(190.3 ± 5.8)	−(179.3 ± 11.8)	−(187.7 ± 9.8)
	RT110				
	RT140				
Alkalinity (mg·L ^{−1})	RTCK	3800 ± 50	4367 ± 102	5067 ± 144	4900 ± 132
		3883 ± 76	5067 ± 152	4800 ± 164	4767 ± 100
	RT50	3766 ± 38	4765 ± 104	5133 ± 76	5216 ± 28
		4150 ± 50	5050 ± 50	5100 ± 138	5100 ± 52
	RT80	3983 ± 29	4683 ± 76	5070 ± 175	5400 ± 78
	RT110				
	RT140				
NH ₄ ⁺ -N Concentration (mg·L ^{−1})	RTCK	305 ± 23	365 ± 12	372 ± 15	565 ± 15
	RT50	332 ± 24	469 ± 16	362 ± 15	486 ± 11
	RT80	363 ± 16	390 ± 15	480 ± 19	584 ± 14
	RT110	506 ± 11	458 ± 14	456 ± 17	673 ± 18
	RT140	449 ± 21	485 ± 11	404 ± 21	493 ± 15
VS removal rate (%)	RTCK	1.8 ± 0.2	3.8 ± 0.4	8.5 ± 0.2	15.0 ± 0.3
	RT50	7.0 ± 0.2	10.2 ± 0.5	13.2 ± 0.4	18.5 ± 0.2
	RT80	4.9 ± 0.4	5.2 ± 0.3	10.0 ± 0.3	19.2 ± 0.4
	RT110	4.9 ± 0.2	4.9 ± 0.4	17.4 ± 0.2	23.7 ± 0.3
	RT140	5.4 ± 0.3	11.2 ± 0.2	18.6 ± 0.4	22.4 ± 0.4

3.2. Performance of methanogenic stage

Under different HTP temperatures, similar trends of daily biomethane production were observed in both one-stage AD and two-stage AD. As shown in Fig. 3A, the daily biomethane production fluctuated between 4.8 ± 0.3 and 24.9 ± 1.1 mL·gVS^{−1} on day 5 to 40 in one-stage AD, following a decreasing trend until to the end of digestion. Moreover, multiple peaks of biomethane production were observed, which might be due to the asynchronous rates of hydrolysis, acidification, and methanation [6]. The results also indicated that HTP had negligible impact on the performance of one-stage AD, corresponding with the findings of Schievano et al. [37]. Comparatively, the daily biomethane production of two-stage AD was significantly enhanced to 6.0 ± 0.3 – 43.1 ± 0.4 mL·gVS^{−1} (Fig. 3B), resulting in the improvement of 25.0–73.1% over that of one-stage AD. Notably, the improvement effect was concentrated from days 10 to 20, which was related to the cumulative VFAs during the DF stage. Similar results were reported by Cheng

et al. [7], who pretreated FW at 160–220 °C. However, a conspicuous lag phase lasted for approximately 4 days during two-stage AD (Fig. 3B), which might be due to the VFAs concentration during the DF stage and toxic byproducts during the HTP process [30].

Overall, a positive correlation was observed between biomethane yield and HTP temperature. In one-stage AD (Fig. 3C), the biomethane yield was increased by 2.8, 4.5, 5.1, and 10.5% in RO50, RO80, RO110, and RO140, respectively, when compared with that of ROCK (435.9 ± 15.0 mL·gVS^{−1}). With HTP (Fig. 3D), the biomethane yields of 549.7 ± 12.8 , 572.4 ± 9.1 , 581.5 ± 10.2 , and 590.6 ± 8.0 mL·gVS^{−1} were observed in RT50, RT80, RT110, and RT140 with an improvement of 4.0, 8.3, 10.1, and 11.8%, respectively, over that of RTCK (528.4 ± 10.8 mL·gVS^{−1}), and were 26.1, 31.3, 33.4, and 35.5% higher than that of ROCK, respectively. Similar results were reported by Xiao et al. [38]. The results also indicated that HTP significantly improved the digestion performance. Under the same duration time, 21.2–26.9% more biomethane was generated in two-stage AD than in one-stage AD. The improvement was primarily owing to the optimal conditions for biomethane generation by methanogens during the methanogenic stage [22]. Overall, the performance and efficiency of the methanogenic stage were significantly enhanced by the HTP.

3.3. Bioenergy efficiency assessment

Similar to previous studies, significant differences in overall energy recovery were observed for one-stage AD and two-stage AD with and without HTP, as shown in Fig. 4. Along with the an increasing HTP temperature, an upward trend was observed in overall BGY, which was 17.28 ± 0.59 – 19.10 ± 0.48 kJ·gVS^{−1} for one-stage AD and 21.18 ± 0.44 – 24.21 ± 0.32 kJ·gVS^{−1} for two-stage AD. Moreover, 4.8–14.3% and 22.6–40.1% more bioenergy was generated in two-stage AD than in RTCK and ROCK, respectively. The results suggest that two-stage AD with properly operated DF stage may consequently exhibit a higher BGY, primarily because of the enhanced biomethane production and associated hydrogen production. This observation corresponded with previous studies [3,24,39], wherein a higher BGY was achieved in two-stage AD, than one-stage AD. Simultaneously, the BCE in two-stage AD was 71.69 ± 1.34 – $80.37 \pm 2.02\%$, which was 14.27 – 16.91% higher than that of one-stage AD ($57.42 \pm 1.46\%$ – $63.46 \pm 1.14\%$). This result corresponded with the fact that the BCE of two-stage AD could be 10% higher than that of one-stage AD [27,39]. Regarding gas quality, the two-stage AD achieved a H₂-rich gas (H₂ content of 28.24–42.82%) in the DF reactor and had a higher biomethane yield in the methanogenic reactor with biomethane content of 69.72–78.64%, which was the root cause for the high BGY [4,19,38]. Considering the energy input for the HTP, DF, and methanogenic stages, a decreasing trend in PEE was observed with increasing HTP temperature. The PEE of two-stage AD ranged from $22.31 \pm 0.56\%$ to $39.01 \pm 0.38\%$, which was 24.83% to 31.15% higher than that of one-stage AD ($17.61 \pm 0.58\%$ – $31.48 \pm 0.34\%$). This suggests that the additional bioenergy generated is insufficient to overcome the added energy required for the process. Similar results were reported by Lin et al. [26]. However, if 90% potential heat was recovered from HTP stage, the PEE could be maintained at 30.68–31.48% for one-stage AD and 39.01–40.63% for two-stage AD, indicating that the use of waste heat could be an important factor for PEE. Overall, the combined process involving HTP and two-stage AD of FW and *Enteromorpha* is a promising alternative method for bioenergy generation.

3.4. Microbial community structure

3.4.1. Bacteria sequence during DF stage

The principal component analysis (PCA) could explain approximately 85.34% of the variations in microbial communities, with PC1 and PC2 accounting for 51.53% and 33.81% (Supplementary materials, Fig. S1), respectively. Three clusters of distinct distribution patterns were observed, and only one cluster was formed by R0, indicating that

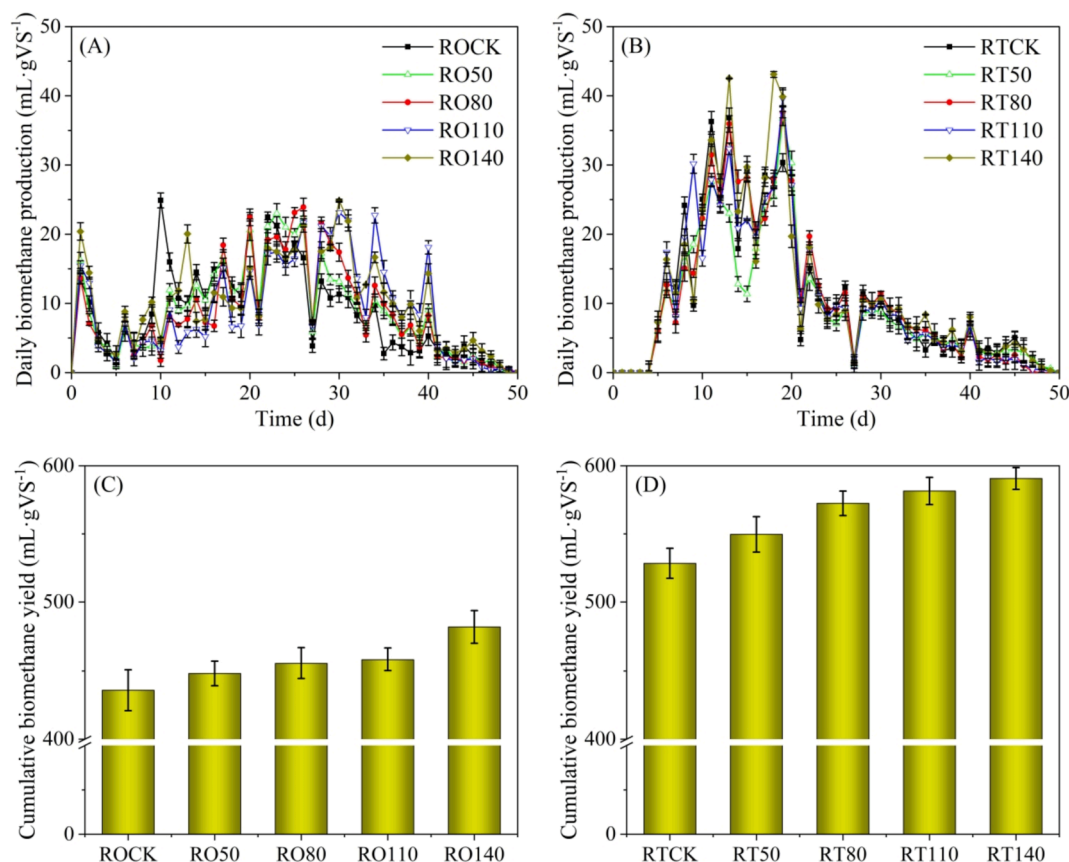


Fig. 3. Daily biomethane production (A, B) and cumulative biomethane yield (C, D) for one-stage and two-stage AD.

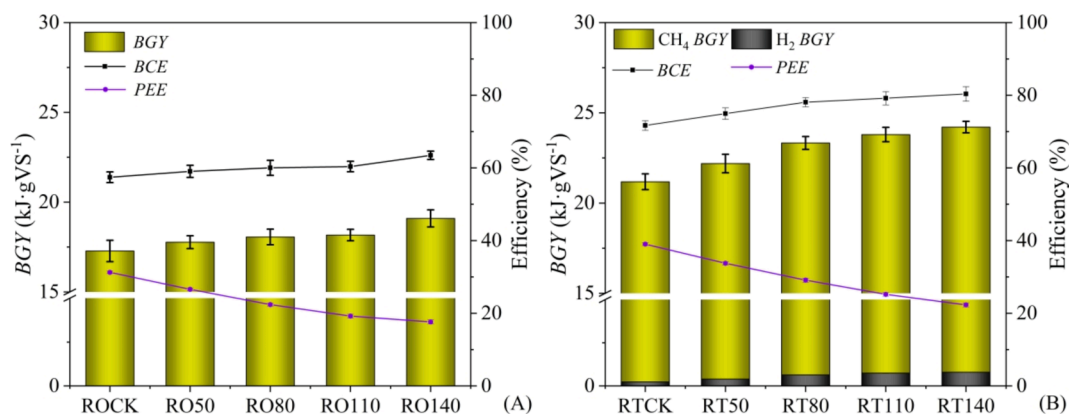


Fig. 4. Bioenergy generation yield (BGY), bioenergy conversion efficiency (BCE), and process energy efficiency (PEE) of one-stage AD (A) and two-stage AD (B).

the microbial community had a clear response to HTP. Moreover, the alpha diversity indices (Supplementary materials, Table S1) of reactors with HTP were considerably higher than those of R0, indicating that the ecological functions of the bacterial community during the DF stage were strengthened by HTP [40].

A community structure comprising the same bacterial phyla was observed in each sample, although their relative abundances varied significantly. *Firmicutes* dominated all reactors with relative abundances of 71.79% in R0 and 77.63–92.84% in RTCK (R1, R2, R3, and R4) and 73.26–92.13% in RT140 (R5, R6, R7, and R8) (Fig. 5A). These results corresponded with those of Koupaie et al. [39], who found that *Firmicutes* was the most dominant phylum with a proportion of 67%–78% in the digestion process coupled with HTP. Moreover, the relative abundance of *Firmicutes* was reduced from 92.13% in R5 to 73.26% in R8, as

well as from 92.84% in R1 to 84.36% in R4. This observation indicates that a negative impact of DF time on the relative abundance of *Firmicutes*. Moreover, the variations in the relative abundance of *Firmicutes* were also related to the substrates, operation parameters, and system stability [21,22]. A similar decreasing trend was observed for *Actinobacteriota*. In contrast, the proportion of *Bacteroidota* was enriched to 1.23–7.17% in RTCK (R1, R2, R3, and R4) and 1.57–4.18% in RT140 (R5, R6, R7, and R8), but was lower in comparison to that (10.16%) of R0. The aforementioned bacteria are typical acid-forming bacteria with the capacity to degrade saccharides, chitin, and protein, and are enriched with high soluble organic matter content [23,35].

The dynamics profiles of the bacteria drastically varied at the genus level, as shown in Fig. 5B. A significant difference ($p < 0.05$) was observed in the major bacterial genera using the chi-square test. In R0,

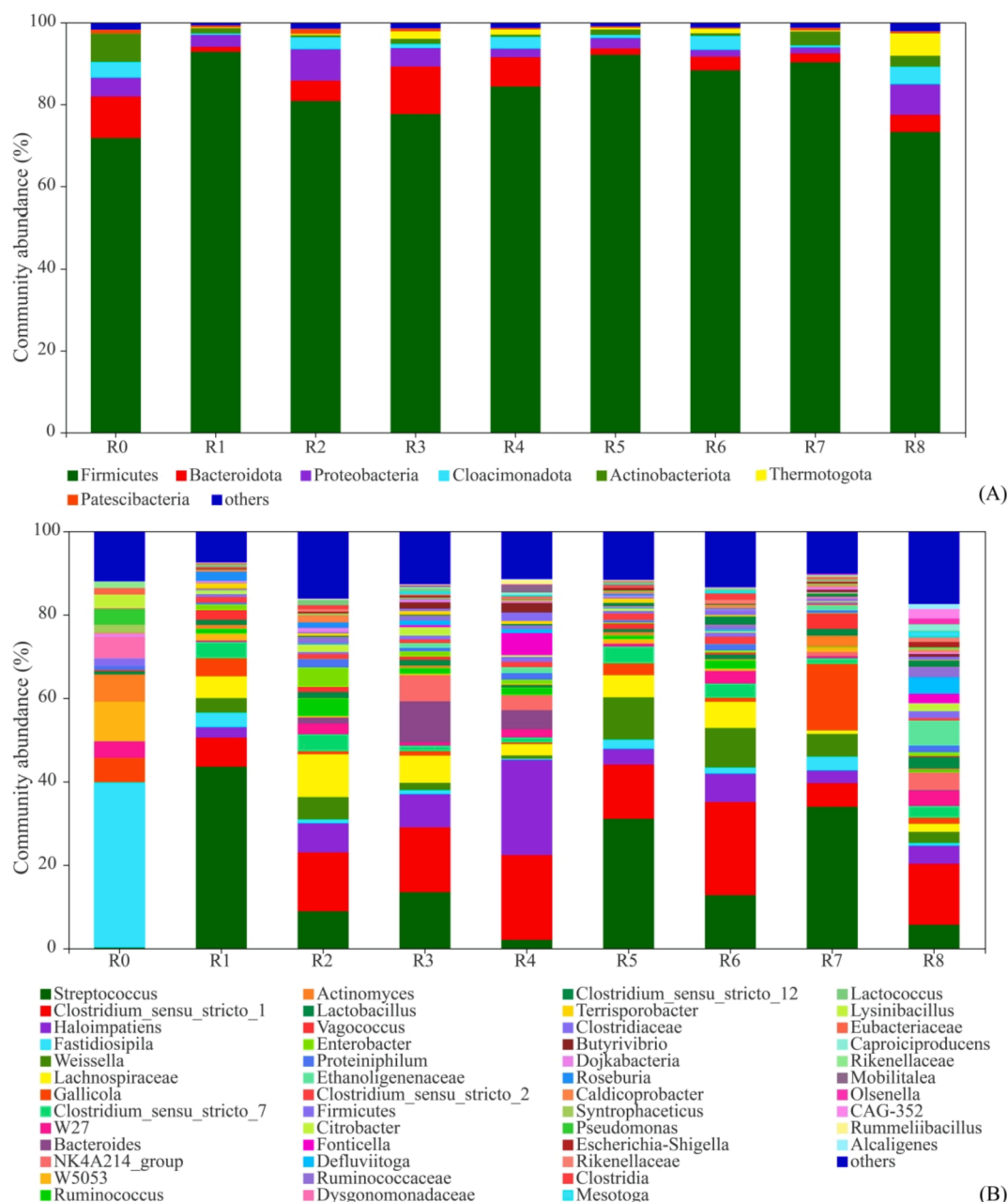


Fig. 5. The relative abundance of major bacteria phylum (a) and genus (b) of inoculum (R0), 12th hour, 24th hour, 4th days, and 7th days of RTCK (R1, R2, R3, and R4, respectively), and RT140 (R5, R6, R7, and R8, respectively) during DF stage.

Fastidiosipila, *W5053*, *Actinomyces*, *Gallicola*, and *Dysgonomonadaceae* were ranked as the five predominant genera with relative abundances of 39.59, 9.41, 6.54, 5.83, and 5.10%, respectively, suggesting a significant difference in bacterial community structure between inoculum and DF of two-stage AD. In RTCK, *Streptococcus*, *Clostridium_sensu_stricto_1*, and *Haloimpatiens* dominated the DF. *Streptococcus* abruptly decreased from 43.56% in R1 to 2.07% in R4, whereas *Clostridium_sensu_stricto_1* and *Haloimpatiens* were increased from 6.97% and 2.51% in R1 to 20.33% and 22.85% in R4, respectively. For RT140, a similar downward trend was observed for *Streptococcus*, which was inhibited from 31.07% in R5 to 5.67% in R8. *Clostridium_sensu_stricto_1* and *Haloimpatiens* fluctuated between 5.67 and 22.33% and 3.03–6.80%, respectively. The aforementioned bacterial genera, as typical hydrogen-producing bacteria, can simultaneously generate VFAs and hydrogen with and without interactions with other bacteria [41,42]. *Clostridium_sensu_stricto_1* was reported to produce higher amounts of hydrogen when compared to

other bacteria; however, its activities were inhibited when extracellular pH was < 5.7 [43,44], which could explain the high hydrogen production yield obtained in RT140 (Fig. 1). The abundance of well-known propionate degraders in all reactors was very low; however, the butyric- and acetic-type fermentation bacteria were found in relatively high abundance in reactors with HTP, which corresponded with the analysis of VFAs. Similar results were reported by Chen et al. [44] and Sun et al. [45]. With an increase in duration, the proportion of *Weissella* decreased from 3.48% in R1 to 0.76% in R4 and from 10.15% in R5 to 2.65% in R8, indicating its sensitivity to the cumulative VFAs concentration. *Gallicola*, frequently reported in hydrogen production fermentation systems, did not exhibit a regular trend in this study, although it exhibited a positive correlation with acetic acid concentration [46]. In addition, some bacteria were enriched in specific samples. More specifically, the largest proportions of *Fastidiosipila*, *Ruminococcus*, *Enterobacter*, *Bacteroides*, *Fonticella*, *Clostridium_sensu_stricto_7*, *Gallicola*, and

Ethanoligenenaceae were found in R1 (3.48%), R2 (4.20%), R2 (4.64%), R3 (9.89%), R4 (5.27%), R5 (4.00%), R7 (15.96%), and R8 (5.97%), respectively, although their proportions were less than 2.00% in the other samples. The abundance of soluble organic matter due to HTP might stimulate the growth of kinetically efficient fermentative bacteria, resulting in varied total VFA concentrations and ethanol as well as bacterial composition, thereby finally enriching or inhibiting specific bacterial genera in turn [21]. Additionally, considering the highest hydrogen and VFAs production in R5 and R7 in RT140, it can be concluded that the ecological functions of bacteria had a direct impact on DF performance rather than the type of bacteria [41]. Consequently, the bacterial community structure in the DF stage exhibited a clear response to HTP.

3.4.2. Archaea sequence during methanogenic stage

The archaeal sequence during the methanogenic stage was also analyzed (Fig. 6). Phylum *Halobacterota* and *Euryarchaeota* dominated in all samples with total relative abundances of 93.44–99.92% (Fig. 6A). The relative abundance of *Euryarchaeota* increased from 17.13% in inoculum (R0) to 13.38–51.50% in one-stage AD (ROCK and RO140) and 21.00–40.88% in two-stage AD (RTCK and RT140), potentially due to faster growth rate of *Euryarchaeota* than that of *Halobacterota* [47]. At the genus level, *Methanosarcina*, *Methanobacterium*, *Methanosaeta*, and

Table 3

Percentage of methanogens (%).

Reactor	Acetoclastic methanogens	Hydrogenotrophic methanogens	Facultative methanogens
Inoculum	0.42	43.85	55.72
ROCK	0.04	16.86	83.11
RTCK	28.92	38.99	32.09
RO140	0.03	56.86	43.12
RT140	63.97	18.39	17.64

Methanoculleus were the most dominant methanogens with significantly varying dynamics profiles. *Methanosarcina* comprised 51.31% of the inoculum, and varied to 37.70–81.46% and 15.65–26.07% in one-stage AD and two-stage AD, respectively (Fig. 6B). Simultaneously, the proportion of *Methanosaeta* was only 0.39% in the inoculum, but enriched up to 23.49% in ROCK and 56.78% in RT140, which conducted under high and low acetate concentrations, respectively [20]. In contrast, *Methanoculleus* was inhibited to 5.04–6.13% and 1.55–5.04% in one-stage AD and two-stage AD with HTP, respectively, from 29.21% in the inoculum. The enrichment of *Methanobacterium* to 28.75% in RTCK and 14.78% in RT140 from 11.17% in the inoculum might have promoted the digestion efficiency of two-stage AD. A similar improvement of the relative abundance of *Methanobrevibacter* was observed in one-stage AD. In addition, *Bathyarchaeia* accounted for a higher proportion in two-stage AD than in one-stage AD, the opposite phenomenon was observed in *Methanoculleus* and *Methanosphaera*. These variations were caused by HTP, which could enhance the methanation process via symbiotic associations with other methanogenic archaea [38,39]. As shown in Table 3, the facultative methanogens and hydrogenotrophic methanogens dominated in the inoculum, accounting for 55.72% and 43.85%, respectively. At an HTP temperature of 140 °C, hydrogenotrophic methanogens were enriched to 56.86% in ROCK, but limited to 18.39% in RT140. Notably, acetoclastic methanogens accounted for 28.92% in RTCK and 63.97% in RT140 with HTP, while the facultative methanogens were inhibited to 32.09 and 17.64%, respectively. The variations in the methanogen percentage might be the reason for the high performance of two-stage AD.

4. Conclusions

In the two-stage AD of FW and *Enteromorpha*, the hydrogen production and acidification performance in the DF stage increased with increasing HTP temperature. The highest hydrogen and biomethane yields of 62.7 ± 0.3 and 590.6 ± 8.0 mL.gVS⁻¹ were obtained at HTP temperature of 140 °C, which were 242.6 and 11.8% higher than those of RTCK, respectively. *BGY* of 21.18 ± 0.44 – 24.21 ± 0.32 kJ.gVS⁻¹ was observed in two-stage AD with HTP. The *BCE* of 71.69–80.37% and *PEE* of 22.31–39.01% in two-stage AD were 14.27–16.91% and 24.83–31.15% higher than those in one-stage AD. Microbial community analysis revealed that butyric- and acetic-type fermentation bacteria were enriched by HTP, and propionate-type fermentation bacteria were inhibited by HTP. Acetoclastic methanogens were enriched by HTP during the methanogenic stage when the temperature was 140 °C. Consequently, HTP could serve as a suitable approach to enhance the efficiency and bioenergy yield of two-stage AD of FW and *Enteromorpha*.

CRedit authorship contribution statement

Yufang Wei: Methodology, Writing – original draft, Investigation, Software. **Jun Gao:** Resources, Investigation. **Zhengui Shi:** Conceptualization, Methodology, Validation. **Xiujin Li:** Supervision, Writing – review & editing. **Wenchao Ma:** Resources, Investigation. **Hairong Yuan:** Supervision, Funding acquisition, Writing – review & editing.

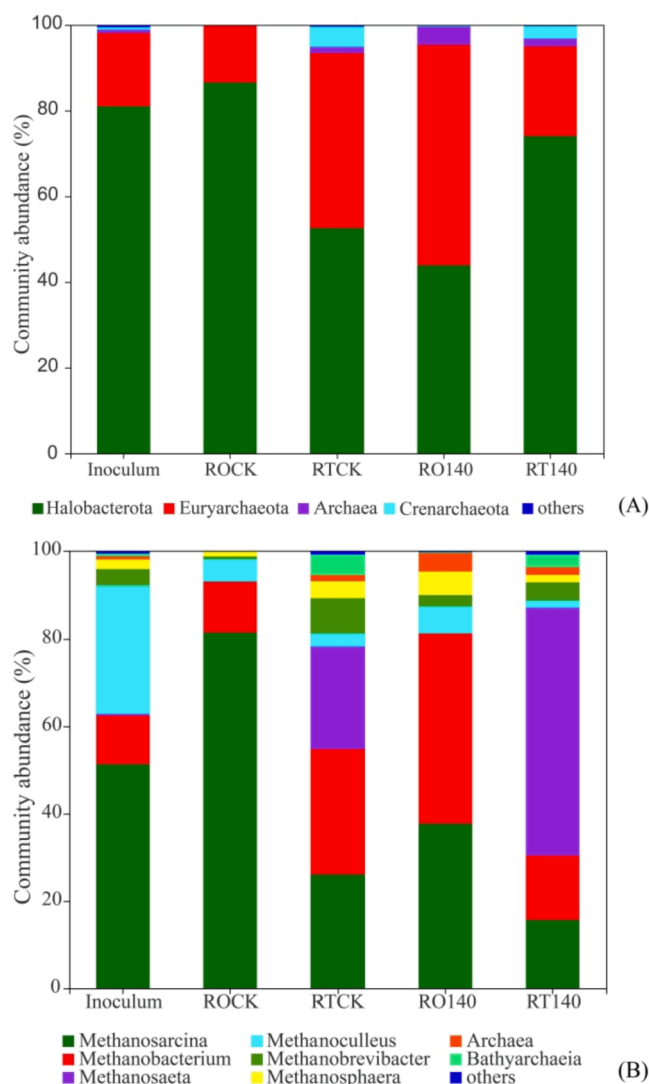


Fig. 6. The relative abundance of major archaea phylum (a) and genus (b) of inoculum, ROCK, RTCK, RO140, and RT140 during the methanogenic stage.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Tianjin City Science and Technology Planning Project (18ZXSZSF00120).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2022.123639>.

References

- [1] Uthirakrishnan U, Godvin Sharmila V, Merrylin J, Adish Kumar S, Dharmadhas JS, Varjani S, et al. Current advances and future outlook on pretreatment techniques to enhance biosolids disintegration and anaerobic digestion: a critical review. *Chemosphere* 2022;288:132553. <https://doi.org/10.1016/j.chemosphere.2021.132553>.
- [2] Jin C, Sun S, Yang D, Sheng W, Ma Y, He W, et al. Anaerobic digestion: an alternative resource treatment option for food waste in China. *Sci Total Environ* 2021;779:146397. <https://doi.org/10.1016/j.scitotenv.2021.146397>.
- [3] De Gioannis G, Muntoni A, Poletti A, Pomi R, Spiga D. Energy recovery from one- and two-stage anaerobic digestion of food waste. *Waste Manage* 2017;68:595–602.
- [4] Baldi F, Pecorini I, Iannelli R. Comparison of single-stage and two-stage anaerobic co-digestion of food waste and activated sludge for hydrogen and methane production. *Renew Energy* 2019;143:1755–65.
- [5] Sun C, Guo L, Zheng Y, Yu D, Jin C, Zhao Y, et al. Effect of mixed primary and secondary sludge for two-stage anaerobic digestion (AD). *Bioresour Technol* 2022;343:126160. <https://doi.org/10.1016/j.biortech.2021.126160>.
- [6] Srisowmeya G, Chakravarthy M, Nandhini Devi G. Critical considerations in two-stage anaerobic digestion of food waste – A review. *Renew Sust Energ Rev* 2020;119:109587. <https://doi.org/10.1016/j.rser.2019.109587>.
- [7] Cheng J, Yue L, Hua J, Dong H, Zhou J, Li Y-Y. Hydrothermal alkali pretreatment contributes to fermentative methane production of a typical lipid from food waste through co-production of hydrogen with methane. *Bioresour Technol* 2020;306:123164. <https://doi.org/10.1016/j.biortech.2020.123164>.
- [8] Mhatre A, Gore S, Mhatre A, Trivedi N, Sharma M, Pandit R, et al. Effect of multiple product extractions on bio-methane potential of marine macrophytic green alga *Ulva lactuca*. *Renew Energy* 2019;132:742–51.
- [9] Chinanews. Qingdao, Shandong welcomes the largest invasion of *Enteromorpha*, 240, 000 tons have been salvaged. 2021–07–05. <https://news.cctv.com/2021/07/05/ARTIQzGd2DLUjxx2EK3JLOfe210705.shtml>.
- [10] Lu J, Liu Z, Zhang Y, Li B, Lu Q, Ma Y, et al. Improved production and quality of biocrude oil from low-lipid high-ash macroalgae *Enteromorpha prolifera* via addition of crude glycerol. *J Clean Prod* 2017;142:749–57.
- [11] Thompson TM, Young BR, Baroutian S. Efficiency of hydrothermal pretreatment on the anaerobic digestion of pelagic *Sargassum* for biogas and fertiliser recovery. *Fuel* 2020;279:118527. <https://doi.org/10.1016/j.fuel.2020.118527>.
- [12] Gao M, Zou H, Tian W, Shi D, Chai H, Gu Li, et al. Co-digestive performance of food waste and hydrothermal pretreated corn cob. *Sci Total Environ* 2021;768:144448. <https://doi.org/10.1016/j.scitotenv.2020.144448>.
- [13] Liu J, Yin J, He X, Chen T, Shen D. Optimizing food waste hydrothermal parameters to reduce Maillard reaction and increase volatile fatty acid production. *J Environ Sci* 2021;103:43–9.
- [14] Song B, Lin R, Lam CH, Wu H, Tsui T-H, Yu Y. Recent advances and challenges of inter-disciplinary biomass valorization by integrating hydrothermal and biological techniques. *Renew Sust Energy Rev* 2021;135:110370. <https://doi.org/10.1016/j.rser.2020.110370>.
- [15] Ding L, Cheng J, Lin R, Deng C, Zhou J, Murphy JD. Improving biohydrogen and biomethane co-production via two-stage dark fermentation and anaerobic digestion of the pretreated seaweed *Laminaria digitata*. *J Clean Prod* 2020;251:119666. <https://doi.org/10.1016/j.jclepro.2019.119666>.
- [16] Ding L, Cheng J, Qiao D, Yue L, Li Y-Y, Zhou J, et al. Investigating hydrothermal pretreatment of food waste for two-stage fermentative hydrogen and methane co-production. *Bioreour Technol* 2017;241:491–9.
- [17] Yin J, Wang K, Yang Y, Shen D, Wang M, Mo H. Improving production of volatile fatty acids from food waste fermentation by hydrothermal pretreatment. *Bioreour Technol* 2014;171:323–9.
- [18] Thompson TM, Young BR, Baroutian S. Advances in the pretreatment of brown macroalgae for biogas production. *Fuel Process Technol* 2019;195:106151. <https://doi.org/10.1016/j.fuproc.2019.106151>.
- [19] Fu S-F, Liu R, Sun W-X, Zhu R, Zou H, Zheng Yi, et al. Enhancing energy recovery from corn straw via two-stage anaerobic digestion with stepwise microaerobic hydrogen fermentation and methanogenesis. *J Clean Prod* 2020;247:119651. <https://doi.org/10.1016/j.jclepro.2019.119651>.
- [20] Wei Y, Li Z, Ran W, Yuan H, Li X. Performance and microbial community dynamics in anaerobic co-digestion of chicken manure and corn stover with different modification methods and trace element supplementation strategy. *Bioreour Technol* 2021;325:124713. <https://doi.org/10.1016/j.biortech.2021.124713>.
- [21] Liao Q, Sun C, Xia Ao, Fu Q, Huang Y, Zhu X, et al. How can hydrothermal treatment impact the performance of continuous two-stage fermentation for hydrogen and methane co-generation? *Int J Hydrogen Energy* 2021;46(27):14045–62.
- [22] Rusín J, Chamrádová K, Basinas P. Two-stage psychrophilic anaerobic digestion of food waste: Comparison to conventional single-stage mesophilic process. *Waste Manage* 2021;119:172–82.
- [23] Shi Z, Zhang L, Yuan H, Li X, Chang Y, Zuo X. Oyster shells improve anaerobic dark fermentation performances of food waste: hydrogen production, acidification performances, and microbial community characteristics. *Bioresour Technol* 2021;335:125268. <https://doi.org/10.1016/j.biortech.2021.125268>.
- [24] Andrade LRS, Cruz IA, Melo L, Vilar D, Fuess LT, Silva GR, et al. Oyster shell-based alkalization and photocatalytic removal of cyanide as low-cost stabilization approaches for enhanced biogas production from cassava starch wastewater. *Process Saf Environ Prot* 2020;139:47–59.
- [25] Zheng H, Sharma A, Ma Q, Zhang C, Hiranuma T, Chen Y, et al. Development of an oyster shell and lignite modified zeolite (OLMZ) fixed bioreactor coupled with intermittent light stimulation for high efficient ammonium-rich anaerobic digestion process. *Chem Eng J* 2020;398:125637.
- [26] Lin RC, Deng C, Ding LK, Bose A, Murphy JD. Improving gaseous biofuel production from seaweed *Saccharina latissima* - The effect of hydrothermal pretreatment on energy efficiency. *Energy Convers Manage* 2019;196:1385–94.
- [27] Yuan T, Bian S, Ko JH, Liu J, Shi X, Xu Q. Exploring the roles of zero-valent iron in two-stage food waste anaerobic digestion. *Waste Manage* 2020;107:91–100.
- [28] Cao Z, Hülsemann B, Wüst D, Illi L, Oechsner H, Kruse A. Valorization of maize silage digestate from two-stage anaerobic digestion by hydrothermal carbonization. *Energy Convers Manage* 2020;222:113218. <https://doi.org/10.1016/j.enconman.2020.113218>.
- [29] Hou T, Zhao J, Lei Z, Shimizu K, Zhang Z. Enhanced energy recovery via separate hydrogen and methane production from two-stage anaerobic digestion of food waste with nanobubble water supplementation. *Sci Total Environ* 2021;761:143234. <https://doi.org/10.1016/j.scitotenv.2020.143234>.
- [30] He C, Hu J, Shen F, Huang M, Zhao Li, Zou J, et al. Tuning hydrothermal pretreatment severity of wheat straw to match energy application scenarios. *Ind Crop Prod* 2022;176:114326. <https://doi.org/10.1016/j.indcrop.2021.114326>.
- [31] Tian W, Chen Y, Shen Y, Zhong C, Gao M, Shi D, et al. Effects of hydrothermal pretreatment on the mono- and co-digestion of waste activated sludge and wheat straw. *Sci Total Environ* 2020;732:139312. <https://doi.org/10.1016/j.scitotenv.2020.139312>.
- [32] Luo L, Kaur G, Zhao J, Zhou J, Xu S, Varjani S, et al. Optimization of water replacement during leachate recirculation for two-phase food waste anaerobic digestion system with off-gas diversion. *Bioreour Technol* 2021;335:125234. <https://doi.org/10.1016/j.biortech.2021.125234>.
- [33] Wang Y, Xi B, Li M, Jia X, Wang X, Xu P, et al. Hydrogen production performance from food waste using piggy anaerobic digested residues inoculum in long-term systems. *Int J Hydrogen Energy* 2020;45(58):33208–17.
- [34] Luo J, Huang W, Zhang Q, Guo W, Xu R, Fang F, et al. A preliminary metatranscriptomic insight of eggshells conditioning on substrates metabolism during food wastes anaerobic fermentation. *Sci Total Environ* 2021;761:143214. <https://doi.org/10.1016/j.scitotenv.2020.143214>.
- [35] Li W, Guo J, Cheng H, Wang W, Dong R. Two-phase anaerobic digestion of municipal solid wastes enhanced by hydrothermal pretreatment: Viability, performance and microbial community evaluation. *Appl Energ* 2017;189:613–22.
- [36] Vesga-Baron A, Etchebehere C, Schiappacasse MC, Chamy R, Tapia-Venegas E. Controlled oxidation-reduction potential on dark fermentative hydrogen production from glycerol: impacts on metabolic pathways and microbial diversity of an acidogenic sludge. *Int J Hydrogen Energy* 2021;46(7):5074–84.
- [37] Schievano A, Tenca A, Scaglia B, Merlino G, Rizzi A, Daffonchio D, et al. Two-stage vs single-stage thermophilic anaerobic digestion: comparison of energy production and biodegradation efficiencies. *Environ Sci Technol* 2012;46(15):8502–10.
- [38] Xiao B, Qin Yu, Wu J, Chen H, Yu P, Liu J, et al. Comparison of single-stage and two-stage thermophilic anaerobic digestion of food waste: Performance, energy balance and reaction process. *Energy Convers Manage* 2018;156:215–23.
- [39] Hosseini Koupaie E, Lin L, Bazyar Lakeh AA, Azizi A, Dhar BR, Hafez H, et al. Performance evaluation and microbial community analysis of mesophilic and thermophilic sludge fermentation processes coupled with thermal hydrolysis. *Renew Sust Energy Rev* 2021;141:110832. <https://doi.org/10.1016/j.rser.2021.110832>.
- [40] Luo L, Wong JWC. Enhanced food waste degradation in integrated two-phase anaerobic digestion: Effect of leachate recirculation ratio. *Bioresour Technol* 2019;291:121813. <https://doi.org/10.1016/j.biortech.2019.121813>.
- [41] Navarro-Díaz M, Valdez-Vázquez I, Escalante AE. Ecological perspectives of hydrogen fermentation by microbial consortia: What we have learned and the way forward. *Int J Hydrogen Energy* 2016;41(39):17297–308.
- [42] Yang G, Wang J. Biohydrogen production from waste activated sludge pretreated by combining sodium citrate with ultrasonic: Energy conversion and microbial community. *Energy Convers Manage* 2020;225:113436. <https://doi.org/10.1016/j.enconman.2020.113436>.
- [43] Castelló E, Nunes Ferraz-Junior AD, Andreani C, Anzola-Rojas MDP, Borzacconi L, Buitrón G, et al. Stability problems in the hydrogen production by dark fermentation: possible causes and solutions. *Renew Sust Energy Rev* 2020;119:109602. <https://doi.org/10.1016/j.rser.2019.109602>.

- [44] Chen Y, Yin Y, Wang J. Influence of butyrate on fermentative hydrogen production and microbial community analysis. *Int J Hydrogen Energy* 2021;46(53):26825–33.
- [45] Sun C, Liao Q, Xia Ao, Fu Q, Huang Y, Zhu X, et al. Degradation and transformation of furfural derivatives from hydrothermal pre-treated algae and lignocellulosic biomass during hydrogen fermentation. *Renew Sust Energy Rev* 2020;131:109983. <https://doi.org/10.1016/j.rser.2020.109983>.
- [46] Yin Y, Wang J. Enhanced biohydrogen production from macroalgae by zero-valent iron nanoparticles: Insights into microbial and metabolites distribution. *Bioresour Technol* 2019;282:110–7.
- [47] Wu S-L, Wei W, Ni B-J. Enhanced methane production from anaerobic digestion of waste activated sludge through preliminary pretreatment using calcium hypochlorite. *J Environ Manage* 2021;295:113346. <https://doi.org/10.1016/j.jenvman.2021.113346>.